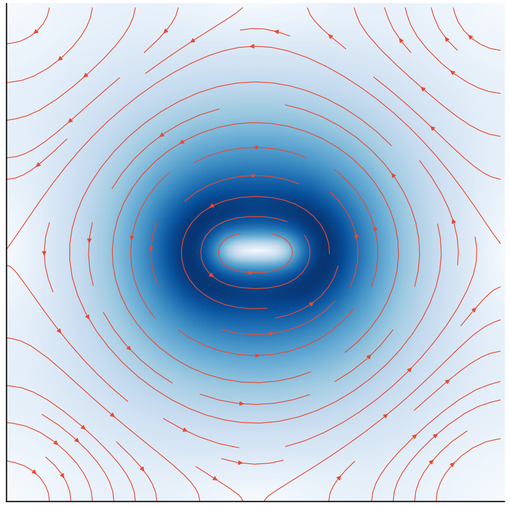
Vortex merger: velocity and streamlines | t = 0.0000 | frame 1/61

Re = 350 | vortex separation = 0.90 | 128 x 128 evaluation grid | identical color scale across all panels

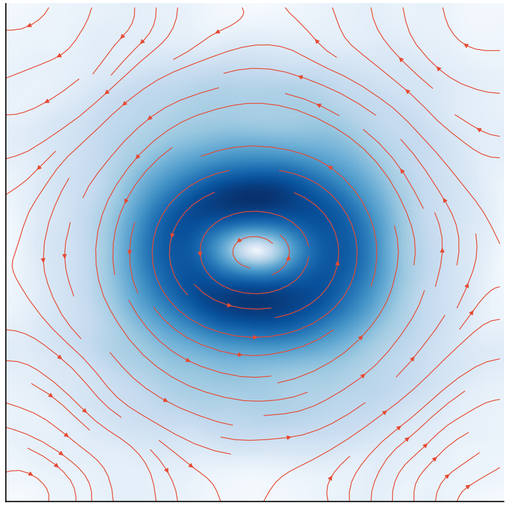
64 modes

GT / DNS



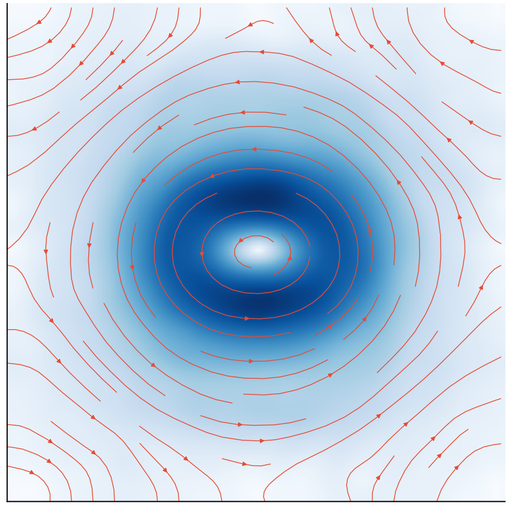
Reference DNS

Gauge-canonical Eigenfluid



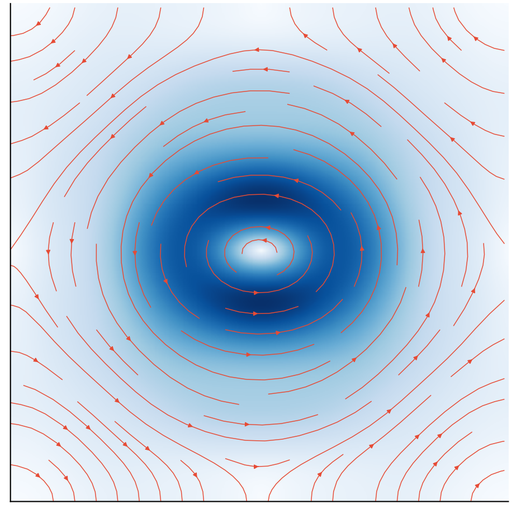
Velocity MSE 2.804e-05 | Vorticity MSE 1.819e-03

Eigenfluid without gauge fixing



Velocity MSE 2.954e-05 | Vorticity MSE 2.044e-03

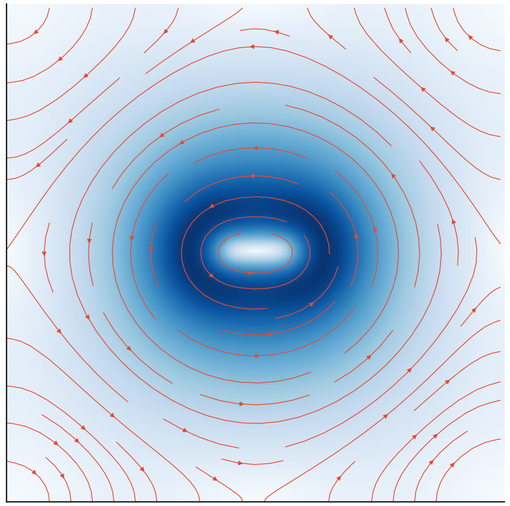
Analytic Laplacian



Velocity MSE 2.541e-05 | Vorticity MSE 1.595e-03

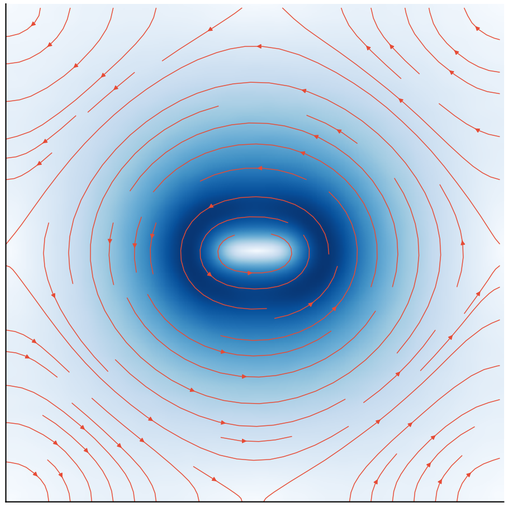
1024 modes

GT / DNS



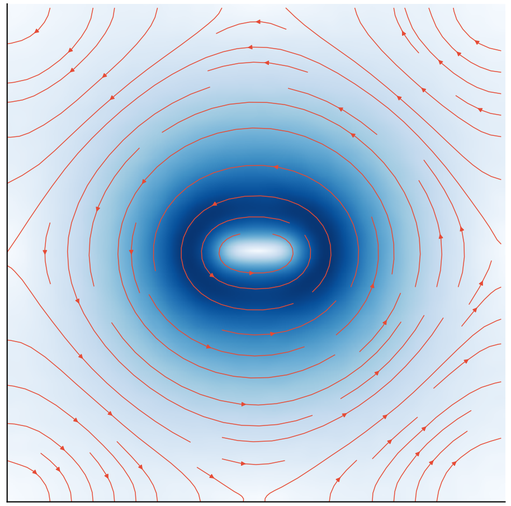
Reference DNS

Gauge-canonical Eigenfluid



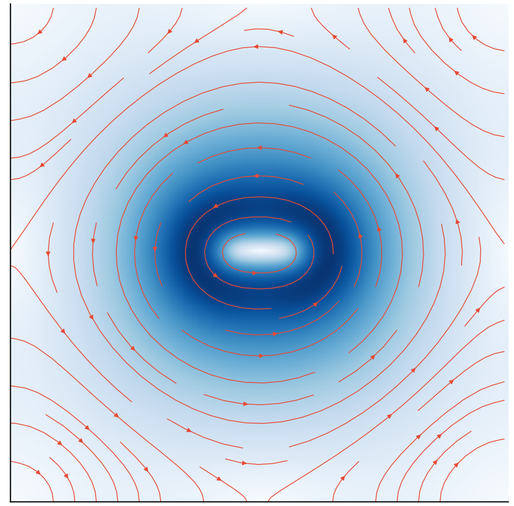
Velocity MSE 1.846e-07 | Vorticity MSE 1.333e-04

Eigenfluid without gauge fixing

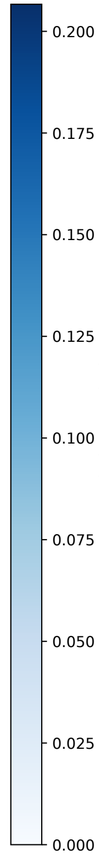


Velocity MSE 3.199e-07 | Vorticity MSE 2.335e-04

Analytic Laplacian



Velocity MSE 1.385e-19 | Vorticity MSE 9.396e-17



Same GT frame 0 projected into each basis; autonomous Navier-Stokes Galerkin rollout thereafter. No later GT refitting.